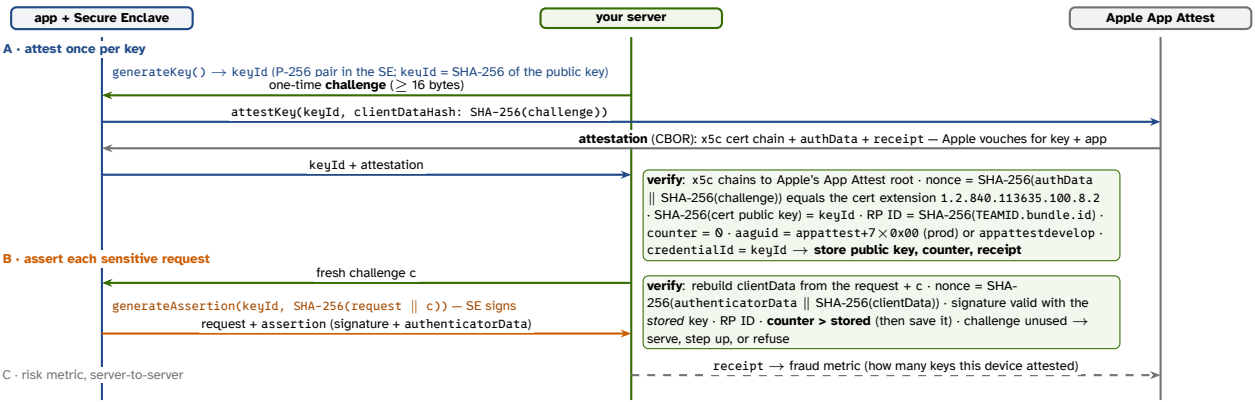


In one line: Attestation lets **your server** ask the *platform vendor* “is this request coming from my genuine, unmodified app on a genuine device?”. **App Attest** makes a Secure-Enclave key, has Apple **attest** it once, then the app **asserts** (signs) each sensitive request with it; **Play Integrity** returns a Google-signed **verdict** about app, device and account. Both are evidence for a server-side risk decision — not authentication, not a jailbreak detector.



App Attest (iOS 14+)

- `DCAppAttestService.shared`: check `isSupported` (false on the Simulator; only action, extensible-SSO and watchOS extensions work even if true). One key **per user per device**; persist the `keyId` — it cannot be recovered.
- `attestKey` calls Apple's server: `serverUnavailable` → retry later with the *same* key; any other error → discard the key, start over. Large apps ramp up gradually (Apple's rollout guidance).
- Keys survive app updates, **not** reinstall, device migration or restore from backup — expect re-attestation, don't flag it as fraud.
- Entitlement `com.apple.developer.devicecheck.appattest-environment = development | production` (the `aaguid` the server must expect).
- Assertions: unlimited per key; bind them to the request (hash of body + challenge), reserve them for sensitive moments. Newer verification steps also check the validation-category and bundle-version extensions in `authData`.
- **DeviceCheck** (`DCDevice.generateToken`): your server asks Apple to read / set **two bits** per device per developer (+ last-update time); they survive reinstall — “promo already used”, not integrity.

Example — client side (the server does all the checking)

```
let svc = DCAppAttestService.shared
guard svc.isSupported else { return .fallback } // other signals
let keyId = try await svc.generateKey() // persist it
let ch = try await api.challenge() // ≥ 16 bytes, one-time
let att = try await svc.attestKey(keyId,
    clientDataHash: Data(SHA256.hash(data: ch)))
try await api.register(keyId: keyId, attestation: att, challenge: ch)
// later, per sensitive request
let body = try JSONEncoder().encode(order); let c = try await api.challenge()
let sig = try await svc.generateAssertion(keyId,
    clientDataHash: Data(SHA256.hash(data: body + c)))
try await api.placeOrder(body, challenge: c, keyId: keyId, assertion: sig)
```

Which one?

	App Attest	DeviceCheck	Play Integrity
proves	genuine app, key in a real SE	nothing: 2 stored bits	app, device, licence, risk
per request	assertion + counter	—	token + requestHash
reinstall	key lost	bits survive	new token per call

Android — Play Integrity API

- **Standard** request: `prepareIntegrityToken` warm-up, then a few hundred ms; `requestHash` = hash of your request; Google mitigates replay. **Classic**: seconds, your own nonce, for rare high-value actions. Default quota 10,000 requests/day.
- **Decrypt on the server**: standard → only via Google's `decodeIntegrityToken`; classic may use your own keys (JWE A256KW/A256GCM wrapping a JWS ES256). Never decrypt in the app.
- **SafetyNet Attestation**: deprecated 2022, fully turned down Jan 2025. Lower level: *Android Key Attestation* (Keystore certificate chain).

<code>appIntegrity</code>	<code>PLAY_RECOGNIZED · UNRECOGNIZED_VERSION · UNEVALUATED</code>
<code>deviceIntegrity</code>	<code>MEETS_BASIC_ / DEVICE_ / STRONG_ / VIRTUAL_INTEGRITY</code> ; empty = hooked, rooted, bad emulator
<code>accountDetails</code>	<code>LICENSED · UNLICENSED</code> (sideloaded)
<code>environmentDetails</code>	<code>appAccessRiskVerdict</code> (capturing / overlay apps), <code>playProtectVerdict</code>
<code>requestDetails</code>	<code>package, requestHash or nonce, timestamp</code>

Android 13+: `MEETS_STRONG_INTEGRITY` also needs a security patch from the last year.

Traps

- ! Checking the attestation *in the app* — the attacker owns that code. Server only.
- ! Skipping the counter or reusing challenges → an old assertion replays.
- ! Hard-blocking when `isSupported` is false or a verdict is missing — locks out real users; tier it: allow, limit, step up, refuse.
- ! A genuine phone in a device farm passes: attestation proves *what* runs, not *who* drives it — add rate limits per key and the fraud metric.
- ! Caching a Play Integrity verdict — Google warns it enables proxying.

Remember

“Attest once, assert often, decide on the server.”